BITS, PILANI – K. K. BIRLA GOA CAMPUS

Database Systems
(CS F212)

by

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Chapter 1: Introduction

- Definition of Database system
- Purpose of Database Systems
- File system versus Database system
- View of Data
- Database Users
- Database Administrator
- Overall System Structure

Database Management System (DBMS)

- DATA vs INFORMATION

- Definition:
  - Collection of interrelated data
  - Set of programs to access the data
  - DBMS contains information about a particular enterprise
  - DBMS provides an environment that is both *convenient* and *efficient* to use.

Database Management System (DBMS)

- Database Applications:
  - Banking: all transactions
  - Airlines: reservations, schedules
  - Universities: registration, grades
  - Sales: customers, products, purchases
  - Manufacturing: production, inventory, orders, supply chain
  - Human resources: employee records, salaries, tax deductions
- Databases touch all aspects of our lives

TYPES OF DATABASES

- Single user or multi user
- Centralized or Distributed
- Structured , semi structured or unstructured

Exercise

- List existing Commercial Database System and its vendors. Also note the type of Database

Purpose of Database System

- In the early days, database applications were built on top of file systems
- Drawbacks of using file systems to store data:
  - Data redundancy and inconsistency
    - Multiple file formats, duplication of information in different files
  - Difficulty in accessing data
    - Need to write a new program to carry out each new task
  - Data isolation — multiple files and formats
  - Integrity problems
    - Integrity constraints (e.g. account balance > 0) become part of program code
    - Hard to add new constraints or change existing ones

Purpose of Database Systems…

- Drawbacks of using file systems (cont.)
  - Atomicity of updates
    - Failures may leave database in an inconsistent state with partial updates carried out
    - E.g. transfer of funds from one account to another should either complete or not happen at all
  - Concurrent access by multiple users
    - Concurrent accessed needed for performance
    - Uncontrolled concurrent accesses can lead to inconsistencies
      - E.g. two people reading a balance and updating it at the same time
  - Security problems
- Database systems offer solutions to all the above problems

File system versus DBS

- Data redundancy and inconsistency
- Difficulty in accessing the data
- Data isolation
- Integrity constraints
- Atomicity
- Concurrency control
- Security

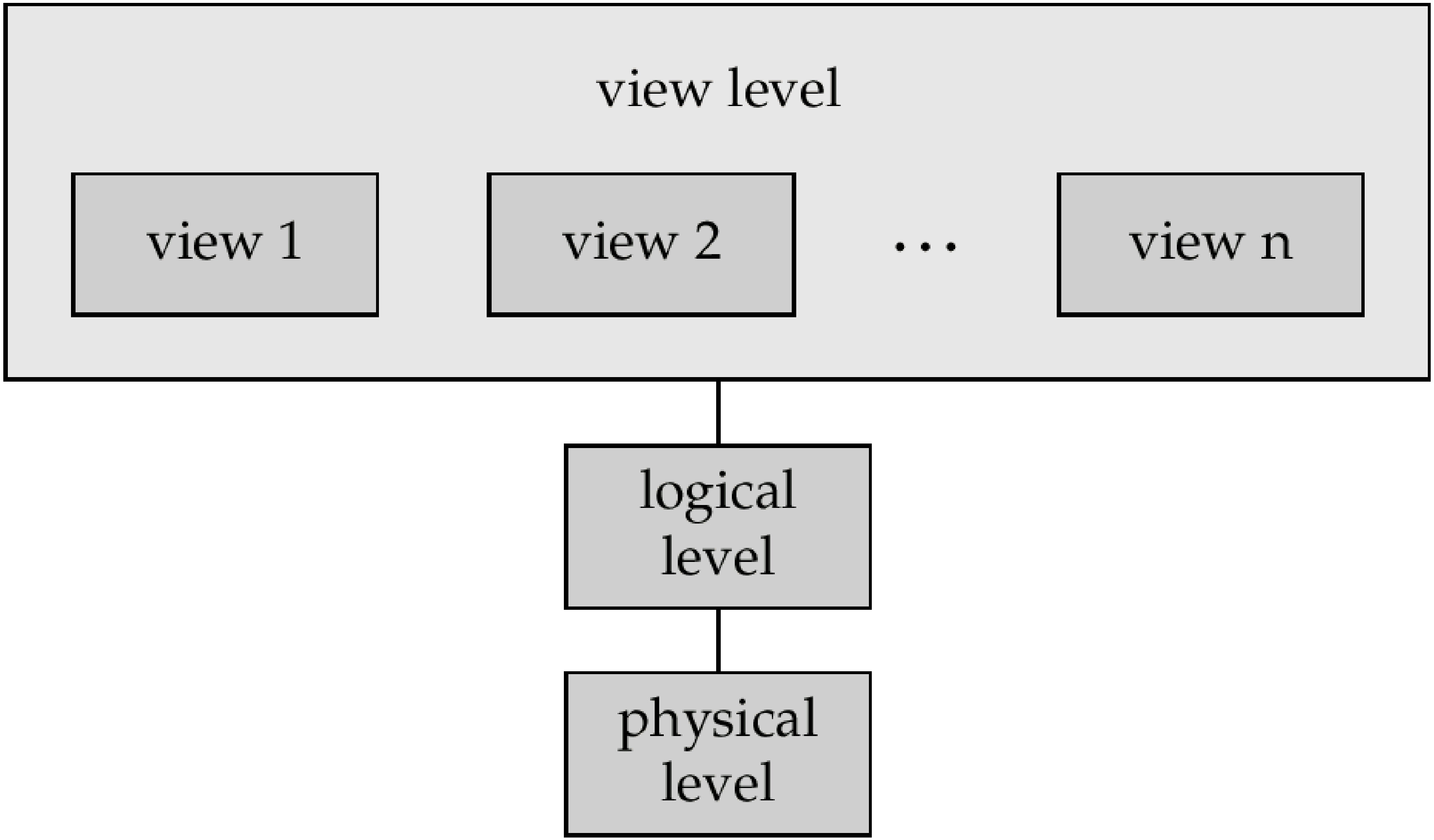
Levels of Abstraction

- Physical level describes how a record (e.g., customer) is stored.
- Logical level: describes data stored in database, and the relationships among the data.

type customer = **record**
name : string;
street : string;
city : integer;
end;

- View level: application programs hide details of data types. Views can also hide information (e.g., salary) for security purposes.

View of Data



Instances and Schemas

- Similar to types and variables in programming language
- **Schema** – the logical structure of the database
  - e.g., the database consists of information about a set of customers and accounts and the relationship between them)
  - Analogous to type information of a variable in a program
  - **Physical schema**: database design at the physical level
  - **Logical schema (Conceptual schema)** : database design at the logical level
- **Instance** – the actual content of the database at a particular point in time
  - Analogous to the value of a variable

Data independence

- **Physical Data Independence** – the ability to modify the physical schema without changing the logical schema
  - Applications depend on the logical schema
  - In general, the interfaces between the various levels and components should be well defined so that changes in some parts do not seriously influence others.

- **Logical Data Independence** – the ability to modify the conceptual schema without changing the external schema or application program.

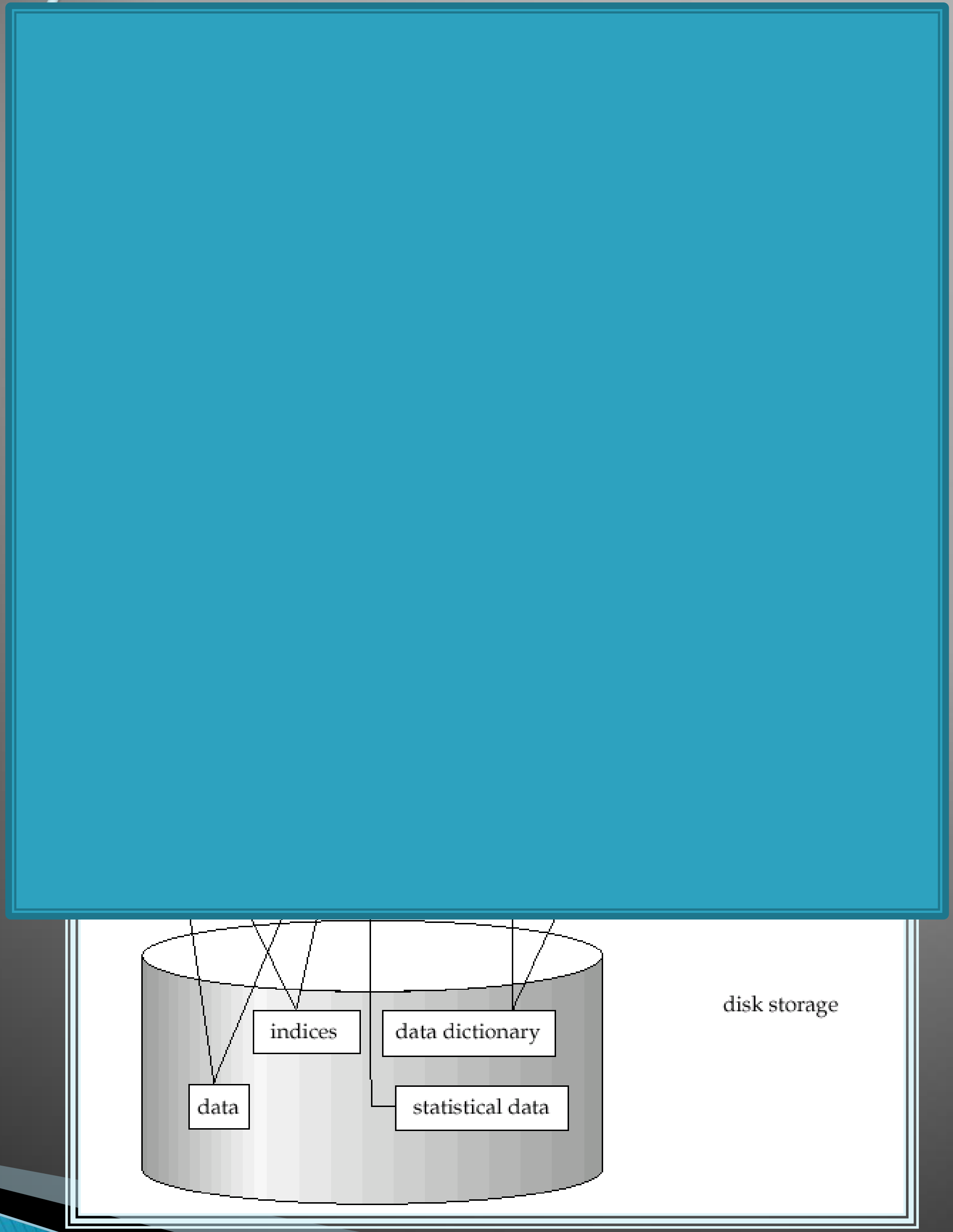
Database Users

- Users are differentiated by the way they expect to interact with the system
- Application programmers – interact with system through DML calls
- Sophisticated users – form requests in a database query language
- Specialized users – write specialized database applications that do not fit into the traditional data processing framework
- Naïve users – invoke one of the permanent application programs that have been written previously
  - E.g. people accessing database over the web, bank tellers, clerical staff

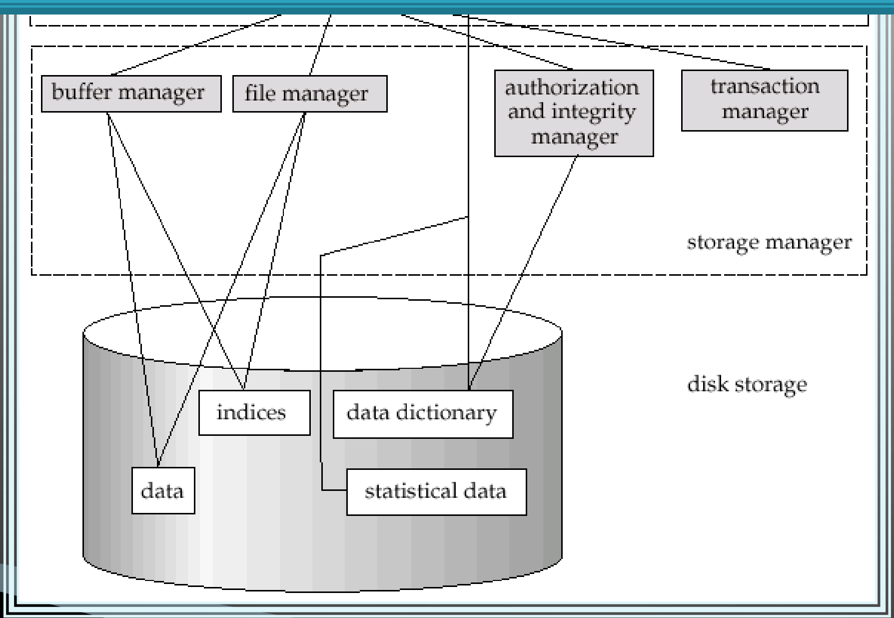
Database Administrator

- Coordinates all the activities of the database system; the database administrator has a good understanding of the enterprise's information resources and needs.
- Database administrator's duties include:
  - Schema definition
  - Storage structure and access method definition
  - Schema and physical organization modification
  - Granting user authority to access the database
  - Specifying integrity constraints
  - Monitoring performance and responding to changes in requirements
  - Periodic backup
  - Recovery of database in case of crash

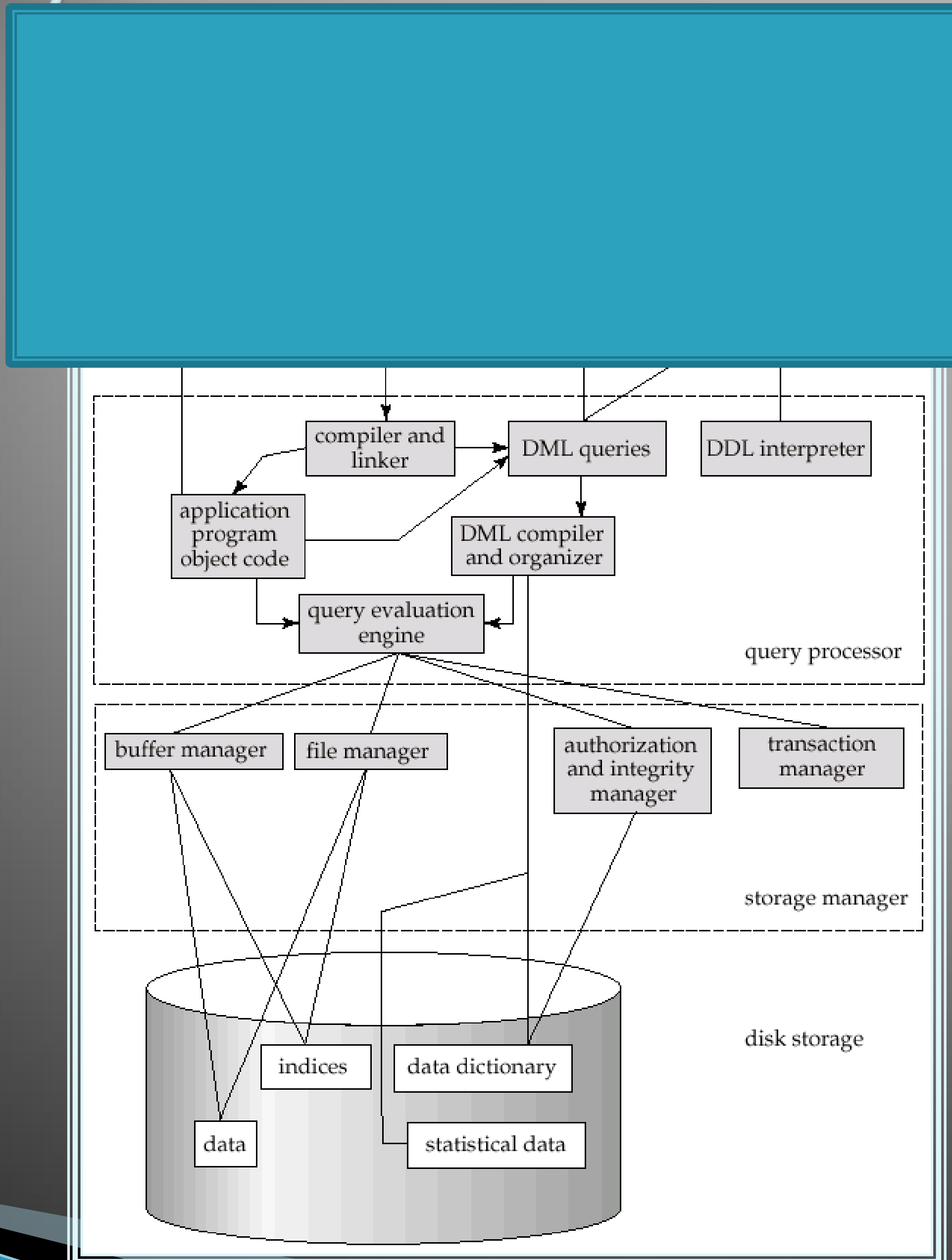
Overall System Structure



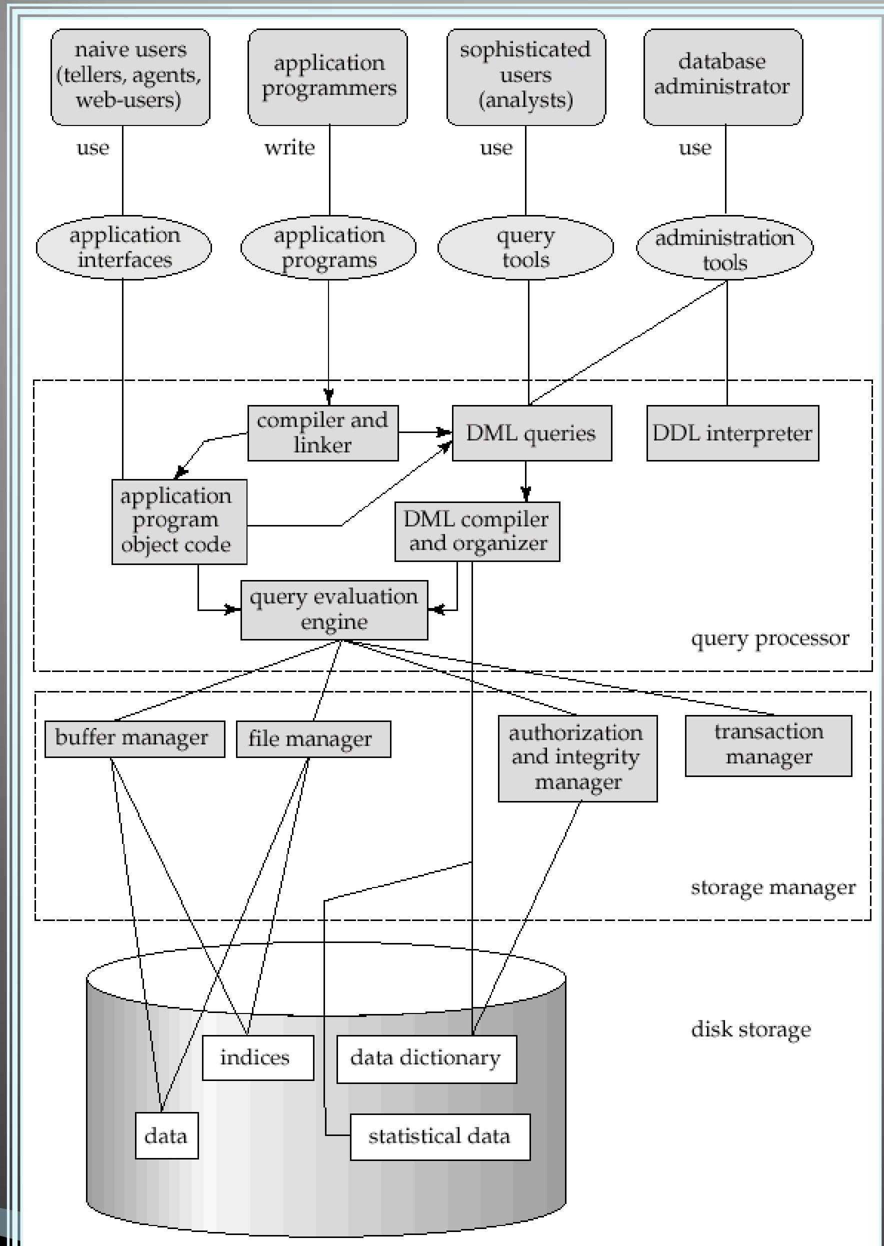
Overall System Structure



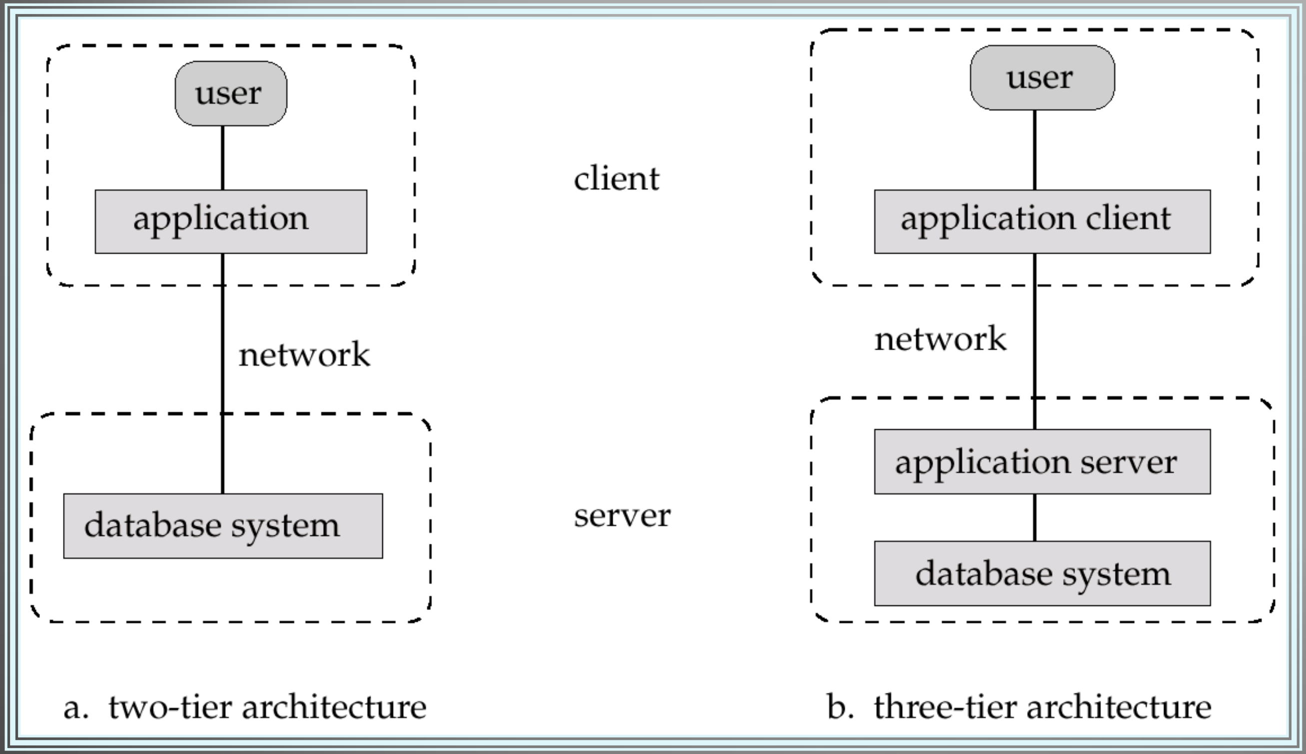
Overall System Structure



Overall System Structure



Application Architectures



- **Two-tier architecture**: E.g. client programs using ODBC/JDBC to communicate with a database
- **Three-tier architecture**: E.g. web-based applications, and applications built using "middleware"